AI-Assisted Fulfilment Reconciliation

Pixi Beauty · Operational Process Proposal · Olayinka Bello

4 hrs/wk	~1 hr/wk	75%	0
Current manual time	Target review time	Time saved	Auto-posts to NetSuite

Process Redesign

CURRENT ~4 hrs/week

01 **Receive files** — Operations
Warehouse CSV + Shopify export downloaded manually each Monday

→ 02 **Manual compare** — Operations
VLOOKUP or side-by-side in Excel; differences flagged by eye

→ 03 **Investigate** — Ops + Finance
Email warehouse for explanations; no structured format

→ 04 **Judgment call** — Finance
Decide exceptions to accept, escalate or chase

→ 05 **ERP prep** — Finance
Manual NetSuite file; error-prone if exceptions unresolved

PROPOSED ~1 hr/week human time

01 **Auto-ingest** — Automated
Shopify webhook + warehouse SFTP parsed each Monday morning

→ 02 **Rule match** — Rules engine
Exact match on ref+SKU+qty; clean cases auto-confirmed instantly

→ 03 **AI triage** — LLM (GPT-4o)
Exceptions classified; warehouse emails parsed; confidence scored

→ 04 **Human review** — Ops lead
Reviews AI suggestions alongside raw evidence; approves/rejects

→ 05 **Approved export** — Finance
NetSuite file only after human sign-off; full audit trail attached

Technology Architecture

SOURCES	BRONZE	SILVER	AI LAYER	GOLD	AUDIT
Data Inputs	**Raw Ingestion**	**Match & Flag**	**Exception Triage**	**Review & Export**	**Traceability**
› Shopify Fulfilment API › Warehouse SFTP/email › Warehouse email replies	› Python ingest script › Timestamped rows › Immutable store	› Normalise order refs › Deterministic engine › Exception rules	› GPT-4o or Claude API › Confidence score 0-1 › Type classification	› Exception dashboard › Approve/Reject/Escalate › Audit log	› AI suggestion stored › Human decision separate › Reviewer + timestamp

GUARDRAIL AI output is stored as a suggestion only. No record moves to Confirmed status without explicit named-human approval. No auto-posting to NetSuite — e

Worked Example — Synthetic Reconciliation Run (week of 14 Oct 2024)

SHOPIFY FULFILMENT EXPORT

Order Ref	SKU	Qty	Value
ORD-4401	SCRB-200ML	3	£74.85
ORD-4402	GLOW-50ML	2	£59.90
ORD-4403	TONER-100ML	5	£99.75
ORD-4403	TONER-100ML	5	£99.75
ORD-4404	MIST-75ML	4	£55.96

WAREHOUSE DISPATCH FILE

Order Ref	SKU	Qty Sent	Dispatch Date
ORD-4401	SCRB-200ML	3	13 Oct
ORD-4402	GLOW-50ML	1	14 Oct
ORD-4403	TONER-100ML	5	14 Oct
ORD-4404	MIST-75ML	4	17 Oct *

* Email: courier hol

Reconciliation Output

Order	Scenario	Match	Handler	Conf.	AI Suggestion	Status
ORD-4401	Clean match	Qty 3=3 ✓	Rules	1.00	Auto-confirmed. No review needed.	CONFIRMED
ORD-4402	Partial ship.	Shopify 2, WH 1	AI	0.72	Likely partial fulfilment — 2nd unit may be back-order.	REVIEW
ORD-4403	Duplicate ref	Ref x2 in Shopify	AI	0.65	Duplicate detected. Warehouse shows 1 dispatch. Do not	ESCALATE
ORD-4404	Email explains	3-day delay	AI	0.91	Warehouse email confirms courier hold. Recommend accept	REVIEW

AI Demo: Exception Explanation Step (runnable Python)

INPUT

```
order_ref: ORD-4402
sku: GLOW-50ML
shopify_qty: 2
warehouse_qty: 1
email: None
```

PROMPT (excerpt)

"Shopify shows qty=2 for ORD-4402 (GLOW-50ML). Warehouse shows qty=1. Classify and suggest an action. Return JSON."

OUTPUT

```
type: "Partial fulfilment"
confidence: 0.72
action: "Confirm 2nd unit
with warehouse before
posting."
escalate: false
```

FAILURE / ESCALATION PATH: If confidence < 0.70, or LLM returns non-JSON, exception is auto-routed to Ops Manager. No action taken without human decision.

6-Week Pilot Plan

Wks 1-2

Shadow run

Pipeline runs alongside manual process. No AI decisions used yet.

Wks 3-4

AI triage, human decisio

Ops lead reviews AI suggestions. Manual process runs as check.

Wk 5

First live export

Finance approves NetSuite file from new system. Manual backup in

Wk 6

Review & decide

Evaluate 4 metrics. Retire manual process if acceptance criteria met.